

# Cloth-Grasp Tilt & Depth Experiment Plan

**Date:** 2026-06-03 **Engineer:** Shiyao Ni **Hardware:** Franka FR3 + arm\_client, fr3\_pose\_controller  
**Contact-force source:** Franka  $F_{\text{ext}}$  (flange, base frame), bias-corrected per run. A301 FlexiForce calibration currently blocked;  $F_{\text{ext}}$  is the first-pass estimate.

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## Objective

Characterize cloth-grasp contact force and tilt tolerance across four gripper variants (rigid vs soft) to quantify the soft-grasper advantage for thin cloth.

## Research questions

1. **Force vs depth** — How does contact force scale with press depth across rigid and soft grippers? (Study 1)
2. **Tilt  $\times$  depth tolerance** — How does tilt about the contact point affect grasp success and contact force, and does this depend on press depth? (Study 2)

## Variables

**Independent** - Gripper variant (4 levels) - Press depth (gripper-specific levels) - Tilt axis  $\in \{x, y, z\}$ , angle  $\in \{0, 2.5^\circ, 5^\circ\}$

**Dependent** - Contact force  $F_{\text{ext}_z}$  (Franka external wrench, base frame, bias-corrected) - Grasp success (visual: cloth held through lift A  $\rightarrow$  B) - Settle (steady-hold) force - Peak press force

**Controlled** - Cloth specimen (single, characterized) - Position A / B in robot base frame - Per-gripper contact-point TCP offset (from touch-off pilot) - Approach / descent / press speeds, settle times - Baseline EEf orientation =  $(180^\circ, 0^\circ, 0^\circ)$  RPY

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## Gripper roster

| ID | Gripper                        | Grasp mode                |
|----|--------------------------------|---------------------------|
| G1 | Parallel jaw, standard (rigid) | force-close, 20 N command |
| G2 | Parallel jaw + TPU 90 A pad    | force-close, 20 N command |
| G3 | Finray, 33 mm                  | fixed gap, 60 mm          |
| G4 | Finray, 18 mm                  | fixed gap, TBD            |

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## Pilot phase (per gripper, one-off)

| #  | Task                       | Method                         | Output                                          |
|----|----------------------------|--------------------------------|-------------------------------------------------|
| P1 | Touch-off at common (x, y) | probe_surface.py, 5 taps @ 2 N | z_table (mean $\pm$ std), TCP offset $\Delta z$ |

| #  | Task                             | Method                                                                 | Output         |
|----|----------------------------------|------------------------------------------------------------------------|----------------|
| P2 | Max-depth escalation, tilt = 0   | step 1 mm until controller reflex (force-safety / accel-discontinuity) | max_depth_safe |
| P3 | Success threshold scan, tilt = 0 | step 1 mm from -1 mm upward; $\geq 2/3$ trials grasp                   | d_min_success  |

$\approx 30$ -40 trials total across grippers.

## Study 1 — depth sweep, tilt = 0

Headline force-vs-depth curve per gripper.

| Gripper | Depths (mm)                                     | Repeats | Trials |
|---------|-------------------------------------------------|---------|--------|
| G1      | -1, 0, 1, 2, 3, 4, 5 (capped by max_depth_safe) | 3       | 21     |
| G2      | -1, 0, 1, 2, 3, 4, 5 (capped)                   | 3       | 21     |
| G3      | -1, 0, 2, 4, 6, 8 (capped)                      | 3       | 18     |
| G4      | -1, 0, 2, 4, 6, 8 (capped)                      | 3       | 18     |

**Study 1 total: 78 trials.**

## Study 2 — tilt $\times$ depth

Tilt about the **per-gripper contact point** (TCP offset from P1). Two depth levels per gripper:

- $d_{\min} = d_{\min\_success}$  — marginal-grasp fragility under tilt
- $d_{\text{safe}} = (d_{\min\_success} + \text{max\_depth\_safe}) / 2$  — comfortable-grasp margin under tilt

Per-gripper tilt conditions (one axis at a time):

| #   | rx (°) | ry (°) | rz (°) | Depth             |
|-----|--------|--------|--------|-------------------|
| T1a | 0      | 0      | 0      | $d_{\min}$        |
| T2a | 2.5    | 0      | 0      | $d_{\min}$        |
| T3a | 5      | 0      | 0      | $d_{\min}$        |
| T4a | 0      | 2.5    | 0      | $d_{\min}$        |
| T5a | 0      | 5      | 0      | $d_{\min}$        |
| T6a | 0      | 0      | 2.5    | $d_{\min}$        |
| T7a | 0      | 0      | 5      | $d_{\min}$        |
| T1b | 0      | 0      | 0      | $d_{\text{safe}}$ |
| T2b | 2.5    | 0      | 0      | $d_{\text{safe}}$ |
| T3b | 5      | 0      | 0      | $d_{\text{safe}}$ |
| T4b | 0      | 2.5    | 0      | $d_{\text{safe}}$ |
| T5b | 0      | 5      | 0      | $d_{\text{safe}}$ |
| T6b | 0      | 0      | 2.5    | $d_{\text{safe}}$ |
| T7b | 0      | 0      | 5      | $d_{\text{safe}}$ |

14 conditions  $\times$  3 reps  $\times$  4 grippers = **168 trials**.

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## Total budget

| Block                           | Trials                                 |
|---------------------------------|----------------------------------------|
| Pilot (P1-P3)                   | ~35                                    |
| Study 1 (depth $\times$ tilt 0) | 78                                     |
| Study 2 (tilt $\times$ depth)   | 168                                    |
| <b>Total</b>                    | <b><math>\approx</math> 280 trials</b> |

Estimated runtime: 3-4 sessions.

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## Open items (to confirm with supervisor)

1. **Gripping direction** in EEf frame (x or y) — determines which tilt axis is *asymmetric* under gripper mirror symmetry. Once confirmed, that axis extends to  $\pm 5^\circ$  (adds ~24 trials).
  2. GRASP\_WIDTH\_M for the **finray 18 mm**.
  3. Whether to **empirically validate** the gripper mirror symmetry on the two “supposed-to-be-symmetric” axes for G1 (full  $\pm 5^\circ$  on those axes — adds 12 trials, provides a “verified” line for the report).
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## Recorded per trial (auto-saved)

- pose + F\_ext CSV at 30 Hz
  - run\_config.json (gripper, depth, tilt, positions, controller settings)
  - config.yaml snapshot
  - Grasp success / failure (manual visual annotation)
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## Symmetry argument (rationale for tilt-level choice)

Each gripper is mirror-symmetric across the plane perpendicular to the gripping direction (the two fingers / blades are identical). Combined with the cloth edge being approximately straight (mirror-symmetric about its perpendicular), two of three tilt axes have  $+\theta \Leftrightarrow -\theta$  outcomes — the rotations whose axes lie *in* the mirror plane. Only the rotation about the gripping direction itself breaks this symmetry (both fingers tilt together  $\rightarrow$  not mirror-equivalent).

Levels  $\{0, 2.5^\circ, 5^\circ\}$  therefore cover the symmetric range. The asymmetric axis (to be identified in item 1 above) will be extended to negative angles in a later patch.

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## Constitutional / governance notes

- **Article II** (human in the loop): all force-driven descents and grasps run with operator present and e-stop in hand; each session is manually authorized.

- **Article I** (signal first): contact force = Franka  $F_{ext\_z}$  (Amendment 1 Signal Inventory row 7), bias-corrected per run. A301 FlexiForce calibration is currently blocked by a non-repeatable raw response (suspected TPU 90 A foot spreading load off the active area);  $F_{ext}$  serves as the first-pass primitive until that is resolved.